

Curriculum Subcommittee Meeting Notes
Thursday, December 10, 2020
Via Zoom

Topic: Review of the MPS Math Curriculum with a presentation delivered by Faiza Khan, Director of Math Department.

Attendance: Paulette Van der Kloot , Chair of Curriculum Subcommittee
Jenny Graham, Member, Curriculum Subcommittee
Paul Ruseau, SC member in place of Mea Mustone

Present: Faiza Khan, Director of Math Department; Dr. Marice Edouard-Vincent, Superintendent; Dr. Peter Cushing, Asst. Superintendent; Secondary, Suzanne Galusi, Asst. Superintendent; Elementary, Rocco Cieri , Director of Science Department; Tom Milaschewski, HGSE Fellow, Bernadette Ricciardelli , Director of Curriculum; Saima A, Cristina McGorty, Michelle Ciccolo, Lisa Evangelista, Alex Lauric, Ingrid Moncada, Meghan Olsen, Ms. Irving, Sharon Hays

4:08: Pledge of Allegiance

Intro: Dr. Marice Edouard-Vincent introduced Faiza Khan

4:10 Presentation: sharing screen

Faiza Khan: outline of presentation

- Common Core, MA Curriculum Frameworks, District Curriculum & District Program

-Any program adopted by the department has been approved by the district and is based on Massachusetts Frameworks (based on Common Core)

Massachusetts has content and practice standards. Practice carries over grade to grade but the content will advance with each year.

The superset of Math Curriculum: MA frameworks which are based on Common Core standards put together by a consortium of states. Massachusetts has added a couple of extra things on top of Common Core. Approved by DESE. Massachusetts frameworks are not tied to specific programs.

Timeline:

Before 1993, states were teaching whatever they wanted (Ed Reform Act in 1993)

1995: MA State Curriculum Frameworks were created in 1995-96

Revisions in November 2000, May 2004, May 2011, December 2017

Medford received most current framework in 2018

Shifts from 2010 are greeted at the higher grades of 6-11

New Grade 8 standards are more rigorous (include concepts from Algebra 1)

2017 Curriculum: well rounded, more word problems (focus on Literacy);
focus on coherence, focus and rigor; conceptual understanding, procedural
fluency and application

K-8 Math Progression Chart (see slide)

- Medford Public Schools Mathematics Department Curriculum & Instruction
2018 - 2020

Math Learning Walks on pause due to COVID; MS program review and
adoption completed; ES program review and adoption on pause; evaluation of
curriculum and instruction are ongoing; inventory of resources is ongoing;
transition to inquiry-based, hands on instructional resources is ongoing;
Summer math calendars were distributed and will be continued (good
success rate in 2019 but not in 2020).

- Elementary Grades

Programming: Envisions 2014 is Core Program, Envisions 2020 supplemental
resource at no cost

Renewed envisions at cost \$50,000

- Secondary Grades
- Middle School

Adoption of new program (Illustrative Mathematics) in June 2019 - emphasis on problem
solving, inquiry based, hands-on activities

Big Ideas - Accelerated Program

Online resources available for families. Work is not necessarily done online. Paper and
pencil are still necessary.

- High School

Programming: Houghton Mifflin Harcourt for Alg. 1, Geo, Alg. 2

Consistency for these three foundational courses. Paid for in 2016

Remaining courses use a variety of publishers

Pearson new Statistics books in Oct. 2020 (\$6000)

HMH Program Renewal due in June 2022. Online components will not be provided after June 2022 if not renewed. Paper books will remain without cost.

Program Review Committee to be formed in June 2022

- Equity in Mathematics
- Future Goals & Next Steps:

Due to COVID, Learning loss in Mathematics predicted to be twice that of Literacy.

COMMENTS:

Paulette Van der Kloot: Key point, what will be the impact of this year? We talk many times of how intense the 8th grade course is? What is the impact of last spring?

Faiza Khan: Teachers are concerned about the decline. Students are not where they should be. Teachers want to make sure the students are engaged. There will have to be a lot of catching up and a lot of below grade level of teaching to be done. I am happy to help them but there is already a very noticeable decline.

Paulette Van der Kloot: diff between remote and hybrid?

Faiza Khan: hard to tell yet. We will be better able to speak about it in February. I am concerned about engagement going down in the winter months. I am hoping teachers will continue working hard with the students and that students wont give up.

Paul Ruseau: Lot to digest here. What about computer science in the list? How young will we go about computer science curriculum?

Faiza Khan: I was teaching AP computer science before taking on this role. A pipeline would be wonderful. What DESE has done recently, is develop digital literacy standards which do not fall under Math. It is fluid right now. Molly Laden has put a lot of effort to get this off the ground. She has introduced a few courses in Comp. Sci. Principles. The course in the Math department is Java based. Molly and I partner with each other.

Jenny Graham: I know the work around Elem. Math was to replace Envisions. I believe that Envisions gets bad marks in the curriculum world. It makes the teachers have to work that much harder to fill the gaps. One of the things we talked about was building a catalog with expiration dates of programs we are offering that way we can make decisions before they expire. We don't want to pay for a program that isn't working. The catalog should also include whether the program includes EL and SPED supplements and whether we have purchased them? As for the HS, the subject of Math progression is an important topic we need to address at a future meeting. Middle school to Calculus pathway? For various textbooks, where are we in the route to renewal? What do we need to be aware of in terms

of budgeting?

Faiza Khan: I agree with much of what you said. A catalog/spreadsheet will be very helpful to give us a place to look in order to plan for the future. If the pandemic hadn't hit, I would have been ready to approach the SC with a plan for an elementary school program purchase. We are postponing for another year. Teachers need a baseline program and one with online access. I believe next March, you will see me for elementary. For high school, we have a good solid program. Statistics is a good book. Advanced Algebra, Trigonometry and Discrete Math might need a purchase down the line. They are not overseen by DESE. We constantly discuss it. We found with the purchase of the Statistics book, the trend is the cost of the book is lower than the cost of a subscription.

Paul Ruseau: are you having conversations with progression at the high school; issues when students move on to next level without adequately finishing previous level (ie. algebra to geometry)

Faiza Khan: Communication is key. I have dept. and individual meetings with teachers about challenges. What didn't help at the high school level, was DESE's announcement that juniors would need to take the MCAS (now delayed). Should teachers stop teaching Algebra 2 and go back to Geometry. It is now not happening in January but it will come back. Might happen in May with the Sophomores. As a department we are discussing it and trying to plan for it. We will need to set time aside to teach to the test. Very unmotivating to the educator. Teachers are all keeping records of where they are in their pacing guides and where they need to take more time. The national council of teachers of mathematics recommend that we fill in the gaps as needed instead of taking time beginning of the year to finish the previous year's material. We have to keep a very realistic eye on where we are. We don't overestimate and need to keep student engagement and connections as a priority.

Paul Ruseau: That makes me happy to hear.

Dr. Marice Edouard-Vincent: The way the middle school team adopted illustrative math and how it was rigorous and engaging to the student. We had to squeeze the training in in June. We planned for Faiza to look at other districts to see what is working but this and the learning walks needed to be put on hold due to COVID. We hope we can start this process for the elementary level in June to improve the programming next year. We hope it will be decided by Spring 2022. The intention is for Secondary gaps to be filled once elementary school is implemented. Since we don't know what is happening yet with the vaccine, we do need to start thinking about next year and if we will have remote learning again in 2021/2022. That will impact our ability to do larger scale instructional learning walks. We are optimistic that we will turn the corner to get back to a greater sense of normalcy.

Jenny Graham: Do any community members have questions?

Faiza Khan: I see a couple of messages to me?

Christina McGorty: Investigations 3?

Faiza Khan: Yes it is on the list?

Lisa: Would like to help with the program review? Formerly on the editorial staff of HMH.

Faiza Khan: We do plan to include parents in the process. Let the teachers look at it first.

Ingrid Moncada (parent): What is the process for choosing an elementary school curriculum? I thought that you had sent out programs you wanted to pilot this year?

Faiza Khan: the five programs under consideration were: Bridges, Investigations, Envisions 2020, Illustrative Math (and one more by Curriculum Associates)
Then we were going to reduce to three then two and then consider the cost. At that point we would recommend it to SC.

Ingrid Moncada: Pilot two?

Faiza Khan: I would like to pilot three but some districts pilot two from the beginning.
No one school with only one program. All three programs will be offered at all the schools. We always recommend two at the end.

Jenny Graham: I know some math curriculums are more dependent on technology. A silver lining of the pandemic might be we are now a one to one device district. That might open the doors to what the students and teachers need. My request is that you don't rule anything out so we can have a conversation about what is needed?

Faiza Khan: Russian Math, Singapore Math and Kumon are great programs but not necessarily aligned with DESE.

Paulette Van der Kloot: has there been any parent feedback? Hirings?

Faiza Khan: new teacher at 7th grade level.

Paulette Van der Kloot; document cameras are delayed for another month or two: do some of the teachers have them?

Faiza Khan: right now, all math classrooms have them. Teachers have good success with them. Good investments.

Paulette Van der Kloot; Circling back to the catalog. Seems like you are well on your way to doing your research. Dr. Marice Edouard-Vincent, would it be appropriate to ask Faiza to provide this information to us?

Jenny Graham: part of the motion is to post it on the website so prospective students can see this information. Name of program/textbook, purchase date, last reviewed date, online access, EL/SPED supplements

Paulette Van der Kloot: It is not the role of the SC to ask you to prepare this document but Superintendent would you like to speak to this?

Dr. Marice Edouard-Vincent: In preparation for today's meeting, there were lots of hyperlinks embedded in the slide deck. As for the course catalog, I have written these fields down and we will address these questions and will particularly address Equity. There are supplemental materials that can assist with EL/SPED as well. Starting with Math, we can follow the same format with the other departments as well. We will definitely be able to complete this document as each department presents.

Faiza Khan: Support for EL and SPED, it is part of our rubric. We look at these factors as we assess programs. ReadyClassroom is the 5th program.

Jenny Graham: Dr. EV, I am sensitive to much work this is for the future . We don't need to hear about the common core and DESE and the history aspect. Focus on where we are right now and what we are considering for the future.

Dr. Marice Edouard-Vincent: I wanted to make people realize that the shift to rigor is fairly recent and why we shifted to Illustrative Math. We can focus on the renewals going forward.

Paulette Van der Kloot: Next meeting focuses on Foreign Language and Technology. Might be a two hour time frame so we can devote an hour to each area. Feb: Science. March: ELA. Jenny asked whether ELA can come sooner.
Thank you for presenting Ms. Khan.

Paul Ruseau: I am excited by this curriculum roadmap. Then the SC can take things in order when it can.

Motion to adjourn: 5:45